THE GEOMETRIC JORDAN PROPERTY FAILS FOR RATIONAL THREEFOLDS

JIHAO LIU

Abstract. We prove that the geometric Jordan property for the Cremona group, conjectured by Moraga, fails in dimension three. More precisely, we construct faithful actions of cyclic groups of arbitrarily large order on smooth rational affine threefolds such that no non-trivial subgroup is contained in an algebraic torus of birational automorphisms.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let n be a positive integer. Prokhorov and Shramov [PS16, Theorem 1.8] proved that, if terminal Fano varieties of dimension n form a bounded family [PS16, Conjecture 1.7], then there exists a positive integer $J(n)$ depending only on n satisfying the following: for any rationally connected variety X of dimension n and any finite subgroup $G \subset \mathrm{Bir}(X)$, there exists a normal abelian subgroup $A \subset G$ with

$$[G : A] \leq J(n).$$

Birkar [Bir21, Theorem 1.1] proved that for any $\epsilon > 0$, the ϵ-lc Fano varieties of dimension n form a bounded family. Since terminal varieties are 1-lc, [Bir21, Theorem 1.1] with $\epsilon = 1$ implies [PS16, Conjecture 1.7], so the conclusion of [PS16, Theorem 1.8] holds unconditionally [Bir21, Corollary 1.5].

Moraga studied how the subgroup A acts geometrically. He proved [Mor24, Theorem 1.1] that there exists a positive integer $M(n)$ depending only on n satisfying the following: for any rationally connected variety X of dimension n and any finite group G acting on X, there exist

(1) a normal abelian subgroup $A \subset G$ with $\operatorname{rank} A \leq n$ and $[G : A] \leq M(n)$,
(2) an A-equivariant birational map $X \dashrightarrow X'$, and
(3) an A-equivariant fibration $X' \to \mathbb{P}^k$ for some non-negative integer k, such that the induced action of A on $\mathbb{P}^k$ factors through $\mathbb{G}_m^k$.

[Mor26, Theorem 8.4] restates [Mor24, Theorem 1.1] for varieties X of Fano type. Moraga adds that it is unclear under which conditions the action of $\mathbb{G}_m^k$ on $\mathbb{P}^k$ lifts to an action of an algebraic torus on X' [Mor24, Section 1.1].

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In [Mor26, Section 8], Moraga proposed the following conjecture and called it the geometric Jordan property for the Cremona group.

Conjecture 1.1 ([Mor26, Conjecture 8.5])**.** *Let n be a positive integer. Then there exists a positive real number $c(n)$ depending only on n satisfying the following. Let X be a rational variety of dimension n and let G be a finite group acting on X. Then there exists a normal abelian subgroup $A \subset G$ such that $[G : A] \leq c(n)$ and A is contained in an algebraic torus in* $\mathrm{Bir}(X)$.

In [Mor26, Conjecture 8.5] the last condition reads $A \subset \mathbb{G}_m^k \subset \mathrm{Bir}(X)$. Since Moraga remarks after the conjecture that it would imply that the action of $\mathbb{G}_m^k$ on $\mathbb{P}^k$ in his theorem above lifts to some birational model of X' [Mor26, Section 8], we read $A \subset \mathbb{G}_m^k \subset \mathrm{Bir}(X)$ as containment of A in an algebraic torus in $\mathrm{Bir}(X)$ in the sense of Definition 2.3, which allows rational actions of tori: an action of G on X is a group homomorphism $G \to \mathrm{Bir}(X)$, and "A is contained in an algebraic torus in $\mathrm{Bir}(X)$" means that the image of A in $\mathrm{Bir}(X)$ is.

Before the conjecture, Moraga mentions birationally rigid threefolds X_m with $\mathbb{Z}/m\mathbb{Z} \subset \mathrm{Aut}(X_m)$ which admit no algebraic torus action, even birationally [Mor26, Section 8]. Birational rigidity implies non-rationality [Pro18, Section 2], so X_m is not rational and does not contradict Conjecture 1.1.

When $n = 2$, Wang [Wan25, Theorem 1.4] proved that there exists a positive integer J_{W} such that any finite subgroup $G \subset \mathrm{Bir}(\mathbb{P}^2)$ contains abelian subgroups A_1 and A_2 with

$$A_1 \trianglelefteq G, \quad [G : A_1] \leq J_{\mathrm{W}}, \quad \text{and} \quad [G : A_2] \leq 2^{11} \cdot 3^2,$$

where A_1 and A_2 are each contained in an algebraic torus of $\mathrm{Bir}(\mathbb{P}^2)$ in the sense of [Wan25]. In the terms of [Wan25, Definition 7.1], the existence of A_1 for every G means that $\mathrm{Bir}(\mathbb{P}^2)$ has the *geometric Jordan property*, and the existence of A_2 for every G means that $\mathrm{Bir}(\mathbb{P}^2)$ has the *weak geometric Jordan property* with bound $2^{11} \cdot 3^2$. In [Mor26], the case $n = 2$ is attributed to Wang and the cases $n \geq 3$ are listed as open.

Non-rational quotients by involutions were used before as an obstruction. Prokhorov [Pro13, Construction 4.5] constructed rational threefolds X with an involution τ such that X/τ is a conic bundle Y over a surface, and showed that τ is not *linearizable*, that is, not conjugate in $\mathrm{Bir}(\mathbb{P}^3)$ to an element of $\mathrm{Aut}(\mathbb{P}^3)$, when Y is not rational. [Pro13, Example 4.6] is such an involution. The obstruction in this paper is also a non-rational quotient by an involution, used against containment in an algebraic torus instead of linearizability.

The goal of this paper is to show that Conjecture 1.1 fails for $n = 3$.

Theorem 1.2. *Conjecture 1.1 does not hold for $n = 3$. More precisely, let $a \geq 4$ be an integer, let $q := 2^{a-1}$, let $h := y^{10} - 1 \in \mathbb{C}[y]$, and let*

$$X_a := \mathrm{Spec}\, \mathbb{C}[x, x^{-1}, y, U, V]/(U^2 - hV^2 - x^{2q} + h^q).$$

Let $G_a \subset \mathbb{C}^*$ be the group of $2q$-th roots of unity, acting on X_a by $\zeta \cdot (x, y, U, V) := (\zeta x, y, U, V)$, let $\sigma \in G_a$ be the element of order 2, and let $\mathbb{C}(X_a)^\sigma \subset \mathbb{C}(X_a)$ be the subfield fixed by σ^*. Then:

1. *X_a is a smooth rational affine threefold, and the action of G_a on X_a is faithful.*
2. *There exist a smooth projective standard conic bundle $\pi : Y \to \mathbb{P}^1_y \times \mathbb{P}^1_z$ (see Definition 2.1) satisfying the following:*
   - (a) *There exists a $\mathbb{C}$-isomorphism $\mathbb{C}(Y) \to \mathbb{C}(X_a)^\sigma$ sending y to y and z to x^2/h.*
   - (b) *The discriminant curve of π has bidegree $(10, q)$.*
   - (c) *The generic fiber $Y_{\mathbb{C}(y,z)}$ of π satisfies*

$$Y_{\mathbb{C}(y,z)} \cong \{u^2 - (y^{10} - 1)v^2 = (z^q - 1)w^2\} \subset \mathbb{P}^2_{\mathbb{C}(y,z)},$$

   where $[u : v : w]$ are homogeneous coordinates on $\mathbb{P}^2_{\mathbb{C}(y,z)}$.
3. *$\mathbb{C}(X_a)^\sigma$ is not purely transcendental over $\mathbb{C}$.*
4. *No non-trivial subgroup of G_a is contained in an algebraic torus in $\mathrm{Bir}(X_a)$ (see Definition 2.3).*

Remark 1.3. Conjugation by a birational map $X_a \dashrightarrow \mathbb{P}^3$, which exists by Theorem 1.2(1), induces an isomorphism $\mathrm{Bir}(X_a) \cong \mathrm{Bir}(\mathbb{P}^3)$ that maps algebraic tori in $\mathrm{Bir}(X_a)$ onto algebraic tori in $\mathrm{Bir}(\mathbb{P}^3)$. More generally, let X' be a variety, let $\varphi : X_a \dashrightarrow X'$ be a birational map, and let ρ be a rational action of an algebraic torus T on X_a with rational map

$$\alpha : T \times X_a \dashrightarrow X_a$$

as in Definition 2.3. For any $t \in T$, since φ restricts to an isomorphism between non-empty open subsets $W \subset X_a$ and $W' \subset X'$, the domain of α intersects $\{t\} \times W$, and $\rho(t)$ is dominant, the rational map

$$\varphi \circ \alpha \circ (\mathrm{id}_T \times \varphi^{-1}) : T \times X' \dashrightarrow X'$$

has domain intersecting $\{t\} \times X'$ and restricts to $\varphi\rho(t)\varphi^{-1}$ on $\{t\} \times X'$. The homomorphism $t \to \varphi\rho(t)\varphi^{-1}$ has kernel $\ker \rho$, and, exchanging the roles of X_a and X', conjugation by φ^{-1} maps algebraic tori in $\mathrm{Bir}(X')$ into algebraic tori in $\mathrm{Bir}(X_a)$. Now let J be a positive integer, take $a \geq 4$ with $2^a > J$, and let $\varphi : X_a \dashrightarrow \mathbb{P}^3$ be a birational map. Any subgroup of $\varphi G_a \varphi^{-1} \subset \mathrm{Bir}(\mathbb{P}^3)$ of index at most J is non-trivial, hence, by Theorem 1.2(4) and the conjugation above, is not contained in an algebraic torus in $\mathrm{Bir}(\mathbb{P}^3)$. Thus $\mathrm{Bir}(\mathbb{P}^3)$ has neither the weak geometric Jordan property nor the geometric Jordan property of [Wan25, Definition 7.1], where algebraic tori are as in Definition 2.3.

Sketch of the proof. Theorem 1.2(1) is Proposition 3.2: for each $\eta \in \mathbb{C}$ such that $\eta^q = 1$, let c_η be a complex number such that $c_\eta^2 = \eta$. Then the polynomials $P, Q \in \mathbb{C}[x, y]$ given by $P + Qw = \prod_{\eta^q=1}(x + c_\eta w)$ in $\mathbb{C}[x, y][w]/(w^2 - h)$ satisfy

$$P^2 - hQ^2 = x^{2q} - h^q,$$

so projecting the conics $\{U^2 - hV^2 = x^{2q} - h^q\}$ from $(U, V) = (P, Q)$ gives a birational map $X_a \dashrightarrow \mathbb{A}^3$.

The functions $z := x^2/h$, $u := U/h^{q/2}$, and $v := V/h^{q/2}$ generate $\mathbb{C}(X_a)^\sigma$ over $\mathbb{C}(y)$ and satisfy

$$u^2 - hv^2 = z^q - 1,$$

so $\mathbb{C}(X_a)^\sigma$ is the function field of a standard conic bundle $\pi : Y \to \mathbb{P}^1 \times \mathbb{P}^1$ whose discriminant curve Δ has bidegree $(10, q)$ (see Construction-Proposition 4.1). Since $4K_{\mathbb{P}^1\times\mathbb{P}^1} + \Delta$ has bidegree $(2, q - 8)$ and $q = 2^{a-1} \geq 8$, Y is not rational (see Sarkisov's criterion, Theorem 2.2). This gives Theorem 1.2(2) and (3).

Since G_a is cyclic of order 2^a, any non-trivial subgroup of G_a contains σ. For any variety X and any element τ of order 2 in an algebraic torus in $\mathrm{Bir}(X)$, we have

$$\mathbb{C}(X)^\tau \cong \mathbb{C}(X)$$

over $\mathbb{C}$ (see Proposition 5.1). Together with Theorem 1.2(1) and (3), this gives Theorem 1.2(4). For Proposition 5.1, let T be an algebraic torus in $\mathrm{Bir}(X)$ containing τ. We may assume that X is affine and T acts regularly on X (see Popov's theorem, Theorem 2.4). Then $\mathbb{C}(X) = \mathbb{C}(X)^T(f_1, \ldots, f_m)$ for the subfield $\mathbb{C}(X)^T$ of T-invariant functions and some T-eigenfunctions $f_1, \ldots, f_m$, algebraically independent over $\mathbb{C}(X)^T$, such that $\tau^* f_1 = -f_1$ and $\tau^* f_j = f_j$ for $j \geq 2$, so

$$\mathbb{C}(X)^\tau = \mathbb{C}(X)^T(f_1^2, f_2, \ldots, f_m) \cong \mathbb{C}(X).$$

Structure of the paper. Section 2 recalls standard conic bundles and Sarkisov's non-rationality criterion, and rational actions of algebraic tori together with Popov's regularization theorem. Section 3 shows that X_a is a smooth rational threefold on which G_a acts faithfully. Section 4 constructs a standard conic bundle birational to the quotient of X_a by σ and shows that it is not rational. Section 5 shows that, for any variety X, the fixed field of an element of order 2 in an algebraic torus in $\mathrm{Bir}(X)$ is isomorphic to $\mathbb{C}(X)$. Section 6 proves Theorem 1.2.

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2. Preliminaries

2.1. Conic bundles.

Definition 2.1. Let $\pi : X \to Z$ be a proper flat morphism of relative dimension 1 between smooth varieties. We say that π is a *conic bundle* if $-K_X$ is π-ample [Pro18, Definition 3.1]. A conic bundle π is *standard* if π^*D is irreducible for any prime divisor $D \subset Z$ [Pro18, Definition 3.1(c)]. If π is a conic bundle and $\dim Z = 2$, then

$$\{z \in Z \mid \pi^{-1}(z) \text{ is singular}\}$$

is the support of a unique reduced divisor $\Delta \subset Z$ [Pro18, 3.3], called the *discriminant curve* of π.

The following non-rationality criterion is due to Sarkisov [Sar81, Corollary 1.8].

Theorem 2.2 ([Pro18, Theorem 12.2 and Corollary 12.4])**.** *Let $\pi : X \to Z$ be a standard conic bundle over a projective rational surface Z, and let Δ be the discriminant curve of π. Assume that*

$$|4K_Z + \Delta| \neq \emptyset.$$

Then X is not rational.

2.2. Algebraic tori in birational automorphism groups.

Definition 2.3. Let X be a variety and let T be an algebraic torus, that is, $T \cong \mathbb{G}_m^n$ for some integer $n \geq 0$. A *rational action* of T on X is a group homomorphism $\rho : T \to \mathrm{Bir}(X)$ for which there exists a rational map $\alpha : T \times X \dashrightarrow X$ such that for any $t \in T$, the domain of α intersects $\{t\} \times X$ and

$$\alpha|_{\{t\}\times X} = \rho(t).$$

ρ is called *regular* if α is a morphism. An *algebraic torus in* $\mathrm{Bir}(X)$ is a subgroup $\rho(T) \subset \mathrm{Bir}(X)$, where T is an algebraic torus and ρ is a rational action of T on X with $\ker \rho = \{1\}$. If a group G acts faithfully on X by birational maps, then we regard G as a subgroup of $\mathrm{Bir}(X)$.

Let ρ be a rational action of an algebraic torus T on a variety X, and let α be as in Definition 2.3. Then ρ is an action in the sense of [Ros56, Section 1], which is the notion of rational action in [Pop16] (see [Pop16, paragraph 1]), since

(1) $(t, t', x) \to \alpha(t, \alpha(t', x))$ and $(t, t', x) \to \alpha(tt', x)$ are rational maps $T \times T \times X \dashrightarrow X$ which restrict to $\rho(tt')$ on each fiber $\{(t, t')\} \times X$, hence they are equal, and
(2) $(t, x) \to (t, \alpha(t^{-1}, x))$ is a rational inverse of $(t, x) \to (t, \alpha(t, x))$, since both composites restrict to $\rho(t^{-1}) \circ \rho(t) = \rho(t) \circ \rho(t^{-1}) = \mathrm{id}_X$ on each fiber $\{t\} \times X$.

Since T is a connected solvable linear algebraic group, the following theorem is [Pop16, Theorem 2(i)] for the group T.

Theorem 2.4 ([Pop16, Theorem 2(i)])**.** *Let T be an algebraic torus and let ρ be a rational action of T on a variety X. Then there exist an affine variety X', a birational map $\varphi : X \dashrightarrow X'$, and a regular action ρ' of T on X' such that $\rho'(t) = \varphi \circ \rho(t) \circ \varphi^{-1}$ for any $t \in T$.*

3. Rationality of the threefolds

Set-up 3.1. Let a, q, h, X_a, G_a, σ, and $\mathbb{C}(X_a)^\sigma$ be as follows.

(1) $a \geq 4$ is an integer, $q := 2^{a-1}$, and $h := y^{10} - 1 \in \mathbb{C}[y]$.
(2) $X_a := \mathrm{Spec}\, \mathbb{C}[x, x^{-1}, y, U, V]/(U^2 - hV^2 - x^{2q} + h^q)$.
(3) $G_a \subset \mathbb{C}^*$ is the group of $2q$-th roots of unity, acting on X_a by $\zeta \cdot (x, y, U, V) := (\zeta x, y, U, V)$.

(4) $\sigma \in G_a$ is the element of order 2, and $\mathbb{C}(X_a)^\sigma \subset \mathbb{C}(X_a)$ is the subfield fixed by σ^*.

Proposition 3.2. *Notation and conditions as in Set-up 3.1. Then X_a is a smooth rational affine threefold, and the action of G_a on X_a is faithful.*

Proof. Let $R := x^{2q} - h^q \in \mathbb{C}[x, y]$, $A := \mathbb{C}[x, x^{-1}, y, V]$, and

$$\Phi := U^2 - hV^2 - R \in A[U].$$

Then $\mathbb{C}[X_a] = A[U]/(\Phi)$. Since Φ is monic of degree 2 in U, $\mathbb{C}[X_a]$ is a free A-module with basis $1, U$, so $A \subset \mathbb{C}[X_a]$.

For any $\eta \in \mathbb{C}$ such that $\eta^q = 1$, let c_η be a complex number such that $c_\eta^2 = \eta$. The ring $\mathbb{C}[x, y][w]/(w^2 - h)$ is a free $\mathbb{C}[x, y]$-module with basis $1, w$, and we write

$$\prod_{\eta^q=1} (x + c_\eta w) = P + Qw, \quad P, Q \in \mathbb{C}[x, y].$$

The $\mathbb{C}[x, y]$-algebra automorphism of $\mathbb{C}[x, y][w]/(w^2 - h)$ sending w to $-w$ sends $P + Qw$ to $P - Qw = \prod_{\eta^q=1}(x - c_\eta w)$. Thus

$$P^2 - hQ^2 = (P + Qw)(P - Qw) = \prod_{\eta^q=1} (x^2 - \eta h) = h^q \prod_{\eta^q=1} (x^2/h - \eta) = x^{2q} - h^q = R.$$

Since h is not a square in $\mathbb{C}[x, x^{-1}, y]$, $hV^2 + R$ is not a square in A, so Φ is a prime element of the unique factorization domain $A[U]$, and X_a is an integral hypersurface in $\operatorname{Spec} A[U] \cong \mathbb{G}_m \times \mathbb{A}^3$. Hence X_a is an affine threefold.

Since $\mathbb{C}[X_a]$ is a free A-module with basis $1, U$, $U - P$ is a non-zero element of the domain $\mathbb{C}[X_a]$. Let

$$t := \frac{V - Q}{U - P} \in \mathbb{C}(X_a).$$

Since $U^2 - hV^2 = R = P^2 - hQ^2$ in $\mathbb{C}[X_a]$, we have

$$(U - P)(U + P) = h(V - Q)(V + Q) = ht(U - P)(V + Q).$$

Dividing by $U - P$ and substituting $V + Q = 2Q + t(U - P)$, we get $U + P = 2hQt + ht^2(U - P)$, that is,

$$(1 - ht^2)(U - P) = 2(hQt - P).$$

If $1 - ht^2 = 0$, then $hQt = P$, so $P^2 = hQ^2 \cdot ht^2 = hQ^2$ and $R = P^2 - hQ^2 = 0$, which is impossible as $0 \neq R \in A \subset \mathbb{C}[X_a]$. Thus $1 - ht^2 \neq 0$ and

$$U = P + \frac{2(hQt - P)}{1 - ht^2}, \quad V = Q + \frac{2t(hQt - P)}{1 - ht^2}.$$

Hence

$$\mathbb{C}(X_a) = \mathbb{C}(x, y, U, V) = \mathbb{C}(x, y, t).$$

Since $\dim X_a = 3$, x, y, t are algebraically independent over $\mathbb{C}$, so X_a is rational.

Since $\partial\Phi/\partial x = -2qx^{2q-1}$ is a unit in $\mathbb{C}[X_a]$, X_a is smooth by the Jacobian criterion.

For any $\zeta \in G_a$, $\zeta^{2q} = 1$, so the $\mathbb{C}$-algebra automorphism of $\mathbb{C}[x, x^{-1}, y, U, V]$ sending x to ζx and fixing y, U, V preserves Φ. Thus G_a acts on X_a by automorphisms. If $\zeta \in G_a$ acts trivially, then $\zeta x = x$ in $\mathbb{C}[X_a]$, so $\zeta = 1$ as x is a unit in $\mathbb{C}[X_a]$. Thus the action of G_a on X_a is faithful. The proposition follows. □

4. Non-rationality of the fixed field of the involution

Construction-Proposition 4.1. Notation and conditions as in Set-up 3.1. Let $Z := \mathbb{P}^1 \times \mathbb{P}^1$ with bihomogeneous coordinates $([y_0 : y_1], [z_0 : z_1])$ and affine coordinates $y = y_1/y_0$ and $z = z_1/z_0$, let $\widetilde{h} := y_1^{10} - y_0^{10}$ and $\widetilde{g} := z_1^q - z_0^q$ be the homogenizations of h and $z^q - 1$, and let

$$\mathcal{E} := \mathcal{O}_Z \oplus \mathcal{O}_Z(-5, 0) \oplus \mathcal{O}_Z(0, -q/2).$$

Let $Y \subset \mathbb{P}_Z(\mathcal{E})$ be the zero locus of the quadratic form $W_0^2 - \widetilde{h}W_1^2 - \widetilde{g}W_2^2$, where $\mathbb{P}_Z(\mathcal{E})$ parametrizes lines in the fibers of $\mathcal{E}$ and W_0, W_1, W_2 are the projections of $\mathcal{E}$ onto $\mathcal{O}_Z$, $\mathcal{O}_Z(-5, 0)$, and $\mathcal{O}_Z(0, -q/2)$, and let $\pi : Y \to Z$ be the projection. Then:

(1) Y is a smooth projective threefold, and π is a standard conic bundle with discriminant curve $\Delta = \{\widetilde{h}\widetilde{g} = 0\}$ of bidegree $(10, q)$.
(2) The open subset $Y \cap \{y_0 z_0 W_2 \neq 0\}$ is the affine hypersurface $u^2 - hv^2 = z^q - 1$ in the coordinates y, z, $u := W_0/(z_0^{q/2} W_2)$, and $v := y_0^5 W_1/(z_0^{q/2} W_2)$, and there exists a $\mathbb{C}$-isomorphism $\mathbb{C}(Y) \to \mathbb{C}(X_a)^\sigma$ sending y, z, u, v to y, x^2/h, $U/h^{q/2}$, $V/h^{q/2}$.
(3) Y is not rational.

Proof. Let $F := \mathbb{C}(X_a)$. Since $a \geq 4$, $q = 2^{a-1} \geq 8$. In particular q is even and $h^{q/2} \in \mathbb{C}[y]$.

Step 1. In this step we prove that

$$F^\sigma = \mathbb{C}(x^2, y, U, V), \tag{4.1}$$

and that the elements x^2/h, $U/h^{q/2}$, $V/h^{q/2}$ of F^σ satisfy

$$\left(\frac{U}{h^{q/2}}\right)^2 - h\left(\frac{V}{h^{q/2}}\right)^2 = \left(\frac{x^2}{h}\right)^q - 1. \tag{4.2}$$

Since $(x, y, U, V) = (1, 0, 0, 0) \in X_a$ and $h(0) = -1$, $h \neq 0$ in F.

Let $K := \mathbb{C}(x^2, y, U, V) \subset F$. Since σ^* fixes x^2, y, U, V, we have $K \subset F^\sigma$. Since $F = K(x)$ and $x^2 \in K$, $[F : K] \leq 2$. Since $\sigma^* x = -x \neq x$, σ^* is an automorphism of F of order 2, so $[F : F^\sigma] = 2$. Thus (4.1) holds. Dividing the relation $U^2 - hV^2 = x^{2q} - h^q$ by h^q, we obtain (4.2). Since

$$x^2 = h \cdot \frac{x^2}{h}, \quad U = h^{q/2} \cdot \frac{U}{h^{q/2}}, \quad \text{and} \quad V = h^{q/2} \cdot \frac{V}{h^{q/2}},$$

(4.1) implies

$$F^\sigma = \mathbb{C}\left(y, \frac{x^2}{h}, \frac{U}{h^{q/2}}, \frac{V}{h^{q/2}}\right). \tag{4.3}$$

Step 2. In this step we prove (1).

The squares W_1^2 and W_2^2 take values in $\mathcal{O}_Z(-10, 0)$ and $\mathcal{O}_Z(0, -q)$. Since

$$\widetilde{h} \in H^0(Z, \mathcal{O}_Z(10, 0)), \quad \widetilde{g} \in H^0(Z, \mathcal{O}_Z(0, q)),$$

we have

$$W_0^2 - \widetilde{h}W_1^2 - \widetilde{g}W_2^2 \in H^0(Z, \operatorname{Sym}^2 \mathcal{E}^\vee) = H^0\big(\mathbb{P}_Z(\mathcal{E}), \mathcal{O}_{\mathbb{P}_Z(\mathcal{E})}(2)\big).$$

Thus Y is the zero locus of a section of $\mathcal{O}_{\mathbb{P}_Z(\mathcal{E})}(2)$, π is projective, and Y is projective.

For $i, j \in \{0, 1\}$, let $Z_{ij} := \{y_i z_j \neq 0\} \subset Z$, and let

$$\widetilde{h}_i := \frac{\widetilde{h}}{y_i^{10}}, \quad \widetilde{g}_j := \frac{\widetilde{g}}{z_j^q}, \quad W_1' := y_i^5 W_1, \quad \text{and} \quad W_2' := z_j^{q/2} W_2.$$

Then $\widetilde{h}_i$ and $\widetilde{g}_j$ are regular functions on Z_{ij}, W_0, W_1', W_2' are the fiber coordinates of a trivialization $\mathcal{E}|_{Z_{ij}} \cong \mathcal{O}_{Z_{ij}}^{\oplus 3}$, and

$$\pi^{-1}(Z_{ij}) = \{Q_{ij} = 0\} \subset Z_{ij} \times \mathbb{P}^2, \quad \text{where } Q_{ij} := W_0^2 - \widetilde{h}_i W_1'^2 - \widetilde{g}_j W_2'^2 \tag{4.4}$$

and $[W_0 : W_1' : W_2']$ are the homogeneous coordinates of $\mathbb{P}^2$. For $i = j = 0$, in the affine coordinates y, z we have $\widetilde{h}_0 = h$ and $\widetilde{g}_0 = z^q - 1$. For a point $p \in Z$, we choose i, j such that $p \in Z_{ij}$. We have

$$\widetilde{h} = \prod_{c^{10}=1} (y_1 - cy_0) \quad \text{and} \quad \widetilde{g} = \prod_{c^q=1} (z_1 - cz_0).$$

Thus $\widetilde{h}_i$ is a polynomial with 10 simple roots in the affine coordinate y_{1-i}/y_i of the first factor of Z, and $\widetilde{g}_j$ is a polynomial with q simple roots in the affine coordinate z_{1-j}/z_j of the second factor. Hence:

- If $\widetilde{h}_i(p) = 0$, then $d\widetilde{h}_i(p)$ is a non-zero multiple of $d(y_{1-i}/y_i)(p)$.
- If $\widetilde{g}_j(p) = 0$, then $d\widetilde{g}_j(p)$ is a non-zero multiple of $d(z_{1-j}/z_j)(p)$.
- If $\widetilde{h}_i(p) = \widetilde{g}_j(p) = 0$, then $d\widetilde{h}_i(p)$ and $d\widetilde{g}_j(p)$ are linearly independent.

By (4.4), $\pi^{-1}(p)$ is the conic

$$\{W_0^2 - \widetilde{h}_i(p)W_1'^2 - \widetilde{g}_j(p)W_2'^2 = 0\} \subset \mathbb{P}^2,$$

defined by a non-zero quadratic form since the coefficient of W_0^2 is 1. Thus all fibers of π are plane conics and have Hilbert polynomial $2t + 1$ with respect to $\mathcal{O}_{\mathbb{P}_Z(\mathcal{E})}(1)$. Since Z is integral, π is flat of relative dimension 1. Since $y_i(p)z_j(p) \neq 0$, $\pi^{-1}(p)$ is singular if and only if $\widetilde{h}(p)\widetilde{g}(p) = 0$. Since Y is locally defined by one equation in the smooth fourfold $\mathbb{P}_Z(\mathcal{E})$ and contains no fiber of $\mathbb{P}_Z(\mathcal{E}) \to Z$, any irreducible component Y' of Y satisfies $\dim Y' = 3$.

By the Jacobian criterion, if Y were singular at a point (p, w) with $w = [w_0 : w_1 : w_2]$ in the coordinates of (4.4), then $w_0 = \widetilde{h}_i(p)w_1 = \widetilde{g}_j(p)w_2 = 0$ and $w_1^2 d\widetilde{h}_i(p) + w_2^2 d\widetilde{g}_j(p) = 0$, which the three items above exclude. Thus Y is smooth.

We show that Y is irreducible. For a general point $p \in Z$, $\widetilde{h}(p)\widetilde{g}(p) \neq 0$, so $\pi^{-1}(p)$ is a smooth, hence irreducible, conic. Let Y' be an irreducible component of Y. Since $\dim Y' = 3$ and $\dim \pi^{-1}(p) = 1$ for any point $p \in Z$, Y' dominates Z. Thus for a general point $p \in Z$, $\dim(Y' \cap \pi^{-1}(p)) = 1$, so $\pi^{-1}(p) \subset Y'$. Hence any two irreducible components of Y meet. Since Y is smooth, no point of Y is contained in two irreducible components of Y, so Y is irreducible. Therefore Y is a smooth projective threefold.

We show that $-K_Y$ is π-ample. Let ξ be a divisor on $\mathbb{P}_Z(\mathcal{E})$ with

$$\mathcal{O}_{\mathbb{P}_Z(\mathcal{E})}(\xi) \cong \mathcal{O}_{\mathbb{P}_Z(\mathcal{E})}(1).$$

Since $\operatorname{rank} \mathcal{E} = 3$, $K_{\mathbb{P}_Z(\mathcal{E})} + 3\xi$ is linearly equivalent to the pullback of a divisor on Z. Since $Y \sim 2\xi$, by adjunction $K_Y + \xi|_Y = (K_{\mathbb{P}_Z(\mathcal{E})} + Y + \xi)|_Y$ is linearly equivalent to the pullback of a divisor on Z. Since $\xi|_Y$ is π-ample, $-K_Y$ is π-ample. Thus π is a conic bundle.

The points $p \in Z$ such that $\pi^{-1}(p)$ is singular form $\{\widetilde{h}\widetilde{g} = 0\}$. The support of $\{\widetilde{h}\widetilde{g} = 0\}$ consists of the 10 curves $\{y_1 = cy_0\}$ with $c^{10} = 1$ and the q curves $\{z_1 = cz_0\}$ with $c^q = 1$, and the $10 + q$ linear factors $y_1 - cy_0$ and $z_1 - cz_0$ of $\widetilde{h}\widetilde{g}$ are pairwise non-proportional, so the divisor $\{\widetilde{h}\widetilde{g} = 0\}$ is reduced. Thus

$$\Delta = \{\widetilde{h}\widetilde{g} = 0\},$$

which is of bidegree $(10, q)$.

We show that π is standard. Let $D \subset Z$ be a prime divisor. Any irreducible component T of $\pi^* D$ satisfies $\dim T = 2$. If $\pi(T)$ were a point, then $\dim T \leq 1$. Thus T dominates D. Hence it suffices to show that $\pi^* D$ is irreducible over a non-empty open subset of D. By the description of Δ above, one of the following holds.

- $D \not\subset \Delta$. Since π is smooth over $Z \backslash \Delta$ with fibers smooth conics, the restriction of $\pi^* D$ to $\pi^{-1}(D \backslash \Delta)$ is smooth over the irreducible curve $D \backslash \Delta$ with irreducible fibers, hence irreducible.
- $D = \{y_1 = cy_0\}$ with $c^{10} = 1$. By (4.4) with $i = j = 0$, over $D \cap \{y_0 z_0 \neq 0\} \cong \mathbb{A}^1_z$ the divisor $\pi^* D$ is

$$\{W_0^2 - (z^q - 1)W_2'^2 = 0\} \subset \mathbb{A}^1_z \times \mathbb{P}^2.$$

Since z^q-1 has simple roots, it is not the square of a polynomial in z, so $W_0^2-(z^q-1)W_2'^2$ is irreducible in $\mathbb{C}[z,W_0,W_1',W_2']$ and π^*D is irreducible there.

- $D=\{z_1=cz_0\}$ with $c^q=1$. By (4.4) with $i=j=0$, over $D\cap\{y_0z_0\neq 0\}\cong\mathbb{A}^1_y$ the divisor π^*D is $\{W_0^2-hW_1'^2=0\}\subset\mathbb{A}^1_y\times\mathbb{P}^2$. Since $h=y^{10}-1$ has simple roots, it is not the square of a polynomial in y, so $W_0^2-hW_1'^2$ is irreducible in $\mathbb{C}[y,W_0,W_1',W_2']$ and π^*D is irreducible there.

Thus π^*D is irreducible, and π is a standard conic bundle. This proves (1).

Step 3. In this step we prove (2).

Take $i=j=0$ in (4.4). Then $Z_{00}=\{y_0z_0\neq 0\}\cong\mathbb{A}^2$ with coordinates y,z, and

$$u=\frac{W_0}{W_2'},\quad v=\frac{W_1'}{W_2'}.$$

Dividing Q_{00} by $W_2'^2$ on $\{W_2\neq 0\}$, we obtain

$$Y^\circ:=Y\cap\{y_0z_0W_2\neq 0\}=\{u^2-hv^2=z^q-1\}\subset\mathbb{A}^4_{y,z,u,v}.$$

Since $(y,z,u,v)=(0,0,\sqrt{-1},0)\in Y^\circ$, Y° is a non-empty open subset of the irreducible smooth threefold Y. Thus the coordinate ring $\mathbb{C}[Y^\circ]=\mathbb{C}[y,z,u,v]/(u^2-hv^2-z^q+1)$ is a domain of dimension 3 with fraction field $\mathbb{C}(Y)$.

By (4.2), the $\mathbb{C}$-algebra homomorphism $\mathbb{C}[y,z,u,v]\to F^\sigma$ sending y, z, u, v to y, x^2/h, $U/h^{q/2}$, $V/h^{q/2}$ induces a homomorphism $\psi:\mathbb{C}[Y^\circ]\to F^\sigma$. By (4.3), the image of ψ generates F^σ as a field. Since $[F:F^\sigma]=2$ and $\dim X_a=3$ by Proposition 3.2, F^σ has transcendence degree 3 over $\mathbb{C}$. Thus $\dim\mathbb{C}[Y^\circ]/\operatorname{Ker}\psi=3=\dim\mathbb{C}[Y^\circ]$. Since $\mathbb{C}[Y^\circ]$ is a domain, $\operatorname{Ker}\psi=0$. Hence ψ extends to a $\mathbb{C}$-isomorphism $\mathbb{C}(Y)\to F^\sigma$ sending y, z, u, v to y, x^2/h, $U/h^{q/2}$, $V/h^{q/2}$. This proves (2).

Step 4. In this step we prove (3).

Since $\mathcal{O}_Z(K_Z)\cong\mathcal{O}_Z(-2,-2)$, and $\mathcal{O}_Z(\Delta)\cong\mathcal{O}_Z(10,q)$ by (1), we have

$$\mathcal{O}_Z(4K_Z+\Delta)\cong\mathcal{O}_Z(2,q-8).$$

Since $q\geq 8$, we have $q-8\geq 0$, so $|4K_Z+\Delta|\neq\emptyset$. By Theorem 2.2, Y is not rational. □

5. Fixed fields of involutions in algebraic tori

Proposition 5.1. *Let X be a variety and let τ be an element of order 2 in an algebraic torus in $\operatorname{Bir}(X)$. Let $\mathbb{C}(X)^\tau$ be the subfield of $\mathbb{C}(X)$ fixed by τ^*. Then*

$$\mathbb{C}(X)^\tau\cong\mathbb{C}(X)\quad\text{over }\mathbb{C}.$$

Proof. By Definition 2.3, there exist an algebraic torus $T\cong\mathbb{G}_m^n$, a rational action $\rho:T\to\operatorname{Bir}(X)$ with $\ker\rho=\{1\}$, and an element $\tau_0\in T$ such that $\tau=\rho(\tau_0)$. Since $\ker\rho=\{1\}$, $\tau_0^2=1$ and $\tau_0\neq 1$.

By Theorem 2.4, there exist an affine variety X', a birational map $\varphi:X\dashrightarrow X'$, and a regular action ρ' of T on X' such that $\rho'(t)=\varphi\circ\rho(t)\circ\varphi^{-1}$ for any $t\in T$. Possibly replacing (X,ρ,τ) with $(X',\rho',\rho'(\tau_0))$, we may assume that X is affine and ρ is regular.

Let $F:=\mathbb{C}(X)$ and let $\widehat{T}:=\operatorname{Hom}(T,\mathbb{G}_m)\cong\mathbb{Z}^n$ be the character group of T, written additively. For any $\lambda\in\widehat{T}$, let

$$F_\lambda:=\{g\in F\mid\rho(t)^*g=\lambda(t)g\text{ for any }t\in T\}.$$

Then $F_0=F^T$, the subfield of F fixed by $\rho(t)^*$ for any $t\in T$, and $F_\lambda F_\mu\subset F_{\lambda+\mu}$ for any $\lambda,\mu\in\widehat{T}$.

Since X is affine and ρ is regular, any $g\in\mathbb{C}[X]$ is a finite sum of elements of the F_λ [Bor91, 1.9 and 8.4].

Let $\Lambda:=\{\lambda\in\widehat{T}\mid F_\lambda\neq 0\}$. Since $1\in F_0$, $F_\lambda F_\mu\subset F_{\lambda+\mu}$, and $g^{-1}\in F_{-\lambda}$ for any non-zero $g\in F_\lambda$, Λ is a subgroup of $\widehat{T}\cong\mathbb{Z}^n$. Thus Λ is free abelian of some rank $m\leq n$. Let $\lambda_1,\dots,\lambda_m$ be a basis of Λ, and let $f_i\in F_{\lambda_i}$ be non-zero for each i. For any $\lambda=\sum_{i=1}^m e_i\lambda_i\in\Lambda$ with $e_i\in\mathbb{Z}$ and

any $g \in F_\lambda$, we have $g\prod_{i=1}^m f_i^{-e_i} \in F_0 = F^T$. By the previous paragraph, $\mathbb{C}[X] \subset F^T(f_1, \dots, f_m)$. Since F is the fraction field of $\mathbb{C}[X]$,

$$F = F^T(f_1, \dots, f_m).$$

For any $e = (e_1, \dots, e_m) \in \mathbb{N}^m$, the monomial $\prod_{i=1}^m f_i^{e_i}$ is a non-zero element of $F_{\sum_i e_i\lambda_i}$, and distinct e correspond to distinct weights $\sum_i e_i\lambda_i$ since $\lambda_1, \dots, \lambda_m$ are linearly independent. Non-zero elements of F_λ for distinct λ are linearly independent over F^T, so $f_1, \dots, f_m$ are algebraically independent over F^T.

For any $\lambda \in \Lambda$, $\lambda(\tau_0)^2 = \lambda(\tau_0^2) = 1$. Thus $\chi(\lambda) := \lambda(\tau_0)$ defines a homomorphism $\chi : \Lambda \to \{\pm 1\}$, and $\tau^* g = \chi(\lambda)g$ for any $g \in F_\lambda$. If χ were trivial, then τ^* would fix F^T and $f_1, \dots, f_m$, hence F, so $\tau = \mathrm{id}_X$, a contradiction. Possibly reordering indices, we may assume that $\chi(\lambda_1) = -1$. Possibly replacing λ_j with $\lambda_j - \lambda_1$ and f_j with f_j/f_1 for any $j \geq 2$ such that $\chi(\lambda_j) = -1$, which keeps $\lambda_1, \dots, \lambda_m$ a basis of Λ and f_j a non-zero element of F_{λ_j}, we may assume that

$$\tau^* f_1 = -f_1, \quad \tau^* f_j = f_j \text{ for any } j \geq 2, \quad \text{and} \quad \tau^*|_{F^T} = \mathrm{id}_{F^T}.$$

Let $F' := F^T(f_1^2, f_2, \dots, f_m)$. Then $F' \subset F^\tau \subsetneq F$, where $F^\tau \neq F$ since $\tau \neq \mathrm{id}_X$. Since $F = F'(f_1)$ and $f_1^2 \in F'$, $[F : F'] \leq 2$, so

$$F^\tau = F' = F^T(f_1^2, f_2, \dots, f_m).$$

Since $f_1, \dots, f_m$ are algebraically independent over F^T, so are $f_1^2, f_2, \dots, f_m$. Thus the F^T-homomorphism $F = F^T(f_1, \dots, f_m) \to F'$ sending f_1 to f_1^2 and f_j to f_j for any $j \geq 2$ is an isomorphism. In particular, $F^\tau \cong F$ over $\mathbb{C}$. The proposition follows. □

6. Proof of the main theorem

Proof of Theorem 1.2. Notation as in Set-up 3.1. (1) is Proposition 3.2.

(2) Let $Z = \mathbb{P}^1 \times \mathbb{P}^1$, $y = y_1/y_0$, $z = z_1/z_0$, $\mathcal{E}$, W_0, W_1, W_2, and $\pi : Y \to Z$ be as in Construction-Proposition 4.1. By Construction-Proposition 4.1(1), Y is smooth and projective, and π is a standard conic bundle whose discriminant curve has bidegree $(10, q)$. By Construction-Proposition 4.1(2), there exists a $\mathbb{C}$-isomorphism $\mathbb{C}(Y) \to \mathbb{C}(X_a)^\sigma$ sending y to y and z to x^2/h. Over $\{y_0 z_0 \neq 0\} \subset Z$,

$$[u : v : w] := [W_0 : y_0^5 W_1 : z_0^{q/2} W_2]$$

are homogeneous coordinates on the fibers of $\mathbb{P}_Z(\mathcal{E})$. Since $\widetilde{h} = y_0^{10}(y^{10} - 1)$ and $\widetilde{g} = z_0^q(z^q - 1)$, we have

$$W_0^2 - \widetilde{h}W_1^2 - \widetilde{g}W_2^2 = u^2 - (y^{10} - 1)v^2 - (z^q - 1)w^2.$$

Thus the generic fiber $Y_{\mathbb{C}(y,z)}$ of π satisfies

$$Y_{\mathbb{C}(y,z)} \cong \{u^2 - (y^{10} - 1)v^2 = (z^q - 1)w^2\} \subset \mathbb{P}^2_{\mathbb{C}(y,z)}.$$

(3) By (2), $\mathbb{C}(X_a)^\sigma \cong \mathbb{C}(Y)$ over $\mathbb{C}$, and Y is not rational by Construction-Proposition 4.1(3). Thus $\mathbb{C}(X_a)^\sigma$ is not purely transcendental over $\mathbb{C}$.

(4) By (1), G_a acts faithfully on X_a, so $G_a \subset \mathrm{Bir}(X_a)$. Let $A \subset G_a$ be a non-trivial subgroup. Since G_a is cyclic of order $2q = 2^a$, $\sigma \in A$. Suppose that $A \subset \rho(T)$ for some algebraic torus $\rho(T)$ in $\mathrm{Bir}(X_a)$. Then $\sigma \in \rho(T)$ and σ has order 2, so by Proposition 5.1,

$$\mathbb{C}(X_a)^\sigma \cong \mathbb{C}(X_a) \quad \text{over } \mathbb{C}.$$

By (1), X_a is rational, so $\mathbb{C}(X_a)$ is purely transcendental over $\mathbb{C}$, which contradicts (3). Thus (4) holds.

Finally, let c be a positive real number, and let $a \geq 4$ be an integer such that $2^a > c$. Let $A \subset G_a$ be a subgroup such that $[G_a : A] \leq c$. Since

$$[G_a : A] \leq c < 2^a = \#G_a,$$

A is non-trivial, so by (4), A is not contained in any algebraic torus in $\mathrm{Bir}(X_a)$. Since c is arbitrary and X_a is rational of dimension 3 by (1), Conjecture 1.1 does not hold for $n = 3$. □

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School of Mathematical Sciences, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Beijing International Center for Mathematical Research, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China
Email address: liujihao@math.pku.edu.cn